

Background: Affordances, the Interaction Cycle, and UX Design Guidelines

Highlights

- A little background on affordances.
- Confusion over affordances in early HCI/UX.
- How functional affordances fit in with Gibson's ecological view of affordances.
- Examples of how cognitive affordances can be informed by shared conventions.
- Where did UX design guidelines come from?

33.1 THIS IS A REFERENCE CHAPTER

This chapter contains reference material relevant to the other chapters of [Part 7](#). This chapter is not intended to be read through as a regular chapter, but each section is supposed to be read when a reference to it in the main chapters is encountered.

33.2 A LITTLE HISTORY OF THE CONCEPT OF AFFORDANCES

Who “invented” the concept of affordances? Of course, we all know it was Donald Norman (not really).

Donald Norman. Norman was the one who introduced the concept of affordances to HCI and UX, using the term in his 1988 book, *The Psychology of Everyday Things*, a book he later renamed *The Design of Everyday Things*. He was a psychologist and rightly saw that usage and usability were about psychology, but the real point of his book was design. In any case, Norman was not the first to use the term “affordance” itself.

J.J. Gibson. It's always risky to make a claim about who used a particular term first, but we think in this case it was J.J. Gibson, more than a decade earlier than when it emerged in HCI. In 1977, he wrote an article on the theory of affordances (Gibson, 1977), and followed up with a complete book on his ecological approach to visual perception (Gibson, 1979).

As an ecological psychologist, Gibson saw animal perception, especially human perception, tied closely to the setting—to the objects that constitute the environment of the animal and how they offer different capabilities to the animal. So, in this light, an affordance is something in a person's environment that helps them do what they need to while they are living their life.

One of the simplest examples he gave was to think about the hard surface of the ground. If it can support my weight, it affords my standing or sitting on that piece of ground. That was Gibson's notion of affordance.

Gaver. Beyond Gibson and Norman, Gaver (1991) had an influence on our thinking about affordances. Gaver (1991) saw affordances in design as a way of focusing on strengths and weaknesses of technologies with respect to the possibilities they offer to people who use them.

He extended the concepts by showing how complex actions can be described in terms of groups of affordances, sequential in time and/or nested in space, showing how affordances can be revealed over time, with successive user actions, for example, in the multiple actions of a hierarchical drop-down menu. Gaver (1991) defined his own terms somewhat differently from those of Norman or Gibson.

McGrenere and Ho. McGrenere and Ho (2000) were another important influence on our work, but they used yet a different terminology that they then compared to Gaver's vocabulary, demonstrating the need for a richer, more consistent, standard vocabulary. Now, the literature about affordances continues to grow. For example, it was featured prominently in CHI 2012, the annual conference in HCI and UX.

That McGrenere and Ho (2000) also needed to calibrate their terminology against Gaver's further demonstrates the difficulty of discussing these concepts without a structured understanding and a more consistent vocabulary.

In most of the related literature, design of cognitive affordances (whatever they are called in a given paper) is acknowledged to be about design for the cognitive part of usability, ease of use in the form of learnability for new and intermittent users (who need the most help in knowing how to do something). All authors who write about affordances give their own definitions of the concept, but there is scant mention beyond cognitive affordances to, for example, physical affordance design.

Physical affordance

A design feature that helps, aids, supports, facilitates, or enables user physical actions: clicking, touching, pointing, gesturing, and moving things (Section 30.3).

Sensory affordance is neglected even more in the literature. Most other authors include sensory affordance only implicitly and/or lumped in with cognitive affordance rather than featuring it as a separate explicit concept. Thus, when these authors talk about perceiving affordances, including [Gaver's \(1991\)](#) and [McGrenere and Ho's \(2000\)](#) phrase “perceptibility of an affordance,” they are referring (in our terms) to a combination of sensing (e.g., seeing) and understanding physical affordances through sensory affordances and cognitive affordances.

Gaver refers to this same mix of affordances when he says, “People perceive the environment directly in terms of its potential for action.” In contrast, our use of the term “sense” has a markedly narrower orientation toward discerning via sensory inputs such as seeing and hearing.

33.3 CONFUSION OVER AFFORDANCES IN EARLY HCI/UX

Norman rightly thought that this idea of affordance was useful in usability and interaction design. But the problem was that people were using the term very loosely in the HCI literature. The term “affordance” was becoming overused, misused, confused, and abused. As a result, the concept was not working well as part of our common ground to communicate efficiently and precisely.

Preamble to Norman's, 1999 article. And Norman, seeing this confusion, got upset and, in good Norman fashion, he decried this situation in his 1999 article, “I was lurking in the background of a CHI-Web discussion, when I lost all reason: I just couldn't take it anymore. ‘I put an affordance here and an affordance there,’ a participant would say, ‘I wonder if that object affords clicking.’ ‘Affordances this, affordances that...’ ‘No!’ I screamed, and out came this note,” ([Norman, 1999](#)).

In hopes of clarifying the distinctions among types of affordances, Norman came up with terms like “real affordances” and “perceived affordances” and tried to tie them to the concepts of Gibson, with whom he had spent some time in discussion.

However, it seems that his attempt to clarify only muddled the waters. People still didn't get it and continued to use the term “affordance” with little shared understanding.

Hartson's taxonomy of affordances. So, I decided to try my hand at it and wrote a paper using terminology that I thought could help HCI and UX practitioners correctly identify and use the different kinds of affordances ([Hartson, 2003](#)).

I tried to connect affordances to human-centered interaction concepts, to the things we humans do as we interact with any kind of machine—our sensory actions, our cognitive actions, and our physical actions. I also thought it was important to show how you can think of interaction design in terms of the affordances that go with these different kinds of actions. So far, the ideas in this article have held up well over time, and they are the foundation for [Chapter 30](#) on affordances.

33.4 EXAMPLES OF HOW COGNITIVE AFFORDANCES CAN BE INFORMED BY SHARED CULTURAL CONVENTIONS

Mailboxes as a symbolic cognitive affordance. One of your authors, Pardha Pyla, grew up in India, and when he would see early software computer systems that used an icon like this for a mail program, he would wonder, well, “What is that thing?” Most homes in India don’t even have a mailbox, the postman knocks (twice?) on the door and delivers the mail directly to the occupant. Even when places did have an external mailbox, they didn’t *look* like this. So, he had no context for sharing that convention.

Only when Pardha got to the United States did he see these things, and, upon close investigation, realize that these are mailboxes. Only later did he also discover it’s a federal offense to investigate other people’s mailboxes.

Exit Signs in the United States versus Europe. The pictures in [Fig. 33-1](#) show two different kinds of exit signs.

The one on the left has shared meaning in the United States. It is bright red, signifying danger (although maybe an exit should represent safety in the face of danger, but never mind). It has arrows indicating the direction to go.

However, in Europe and some other parts of the world, they use something like the sign on the right-hand side of the figure. Its meaning seems fairly clear on

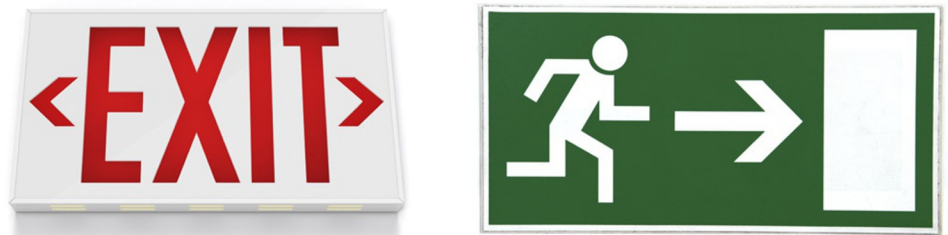


Fig. 33-1
Exit signs from two different cultures.

the surface of it, but in this country, a panicked person in a state of emergency might find it unfamiliar and possibly confusing.

Relying on the inherent characteristics of a Coke bottle. As we said in [Section 30.2.1.4](#), a shared cultural understanding often conveys much of the meaning of cognitive affordances. In the absence of an applicable cultural convention, we are left to rely solely on deducing meaning from the inherent characteristics of an object.

What do you think of the very familiar shape of a Coke bottle? What are some affordances that you can think of that might go with it? Just by looking at it, what do you think you can do with it? Good physical affordance to grip it, the functional affordance to hold liquid to drink, possibly a nice aesthetic look as an emotional affordance. All of these are relative to our own experience in seeing, holding, and using these bottles for their intended purpose—a container for a refreshing drink.

In a graduate class of UX design students, we raised the question of how some of our most reliable cognitive affordances (e.g., how to operate a doorknob) might be perceived by people “from another planet,” who did not have the same shared conventions. We handed out identical empty Coke bottles to several groups and asked them to look at the bottles, to hold and handle them, and to think about what kinds of uses the inherent affordances evoke. We wanted them to get down to Gibson’s ecological level.

Students responded with the usual answers about affordances evinced by sight and touch. Visually, a Coke bottle has obvious affordances as a vessel to hold, for example, flowers, or it can serve as a rough volume-measuring device. The heft and sturdiness sensed when held in one’s hand indicate affordances to serve as a paperweight, a plumb bob on a string, or even an oddly shaped rolling pin.

But what if you’ve never seen or heard of a Coke bottle before? Then how do you figure out what it is and what it is used for? That is the basis for the movie *The Gods Must Be Crazy*.

It opens with the pilot of a small plane flying over Southern Africa. He is drinking the last of a Coke, and he throws the empty bottle out the window. Down and down it floats.

As this puzzling artifact from another culture lands at the feet of a bushman in the deep Kalahari, the scene deals directly with affordances. This man is seeing a Coke bottle for the first time. Naturally, he takes it back to his village to show his friends. To the villagers, it is a strange and beautiful thing, but they are puzzled about what it could be used for (its functional affordances) and how to use it (the cognitive affordances).

Emotional affordance

A design feature that helps a user make an emotional connection resulting in emotional impact within the user experience ([Section 30.6](#)).

Its appearance triggers a curious exploration of the possibilities of this object within their own Kalahari culture. It is transparent, like water, but it was very hard. They wonder why the “gods” have sent this thing down to the earth, and what it might be used for.

This part of the movie makes the point about ecological affordances so well. Because the Kalahari Bushmen have never seen a Coke bottle (or any drink bottle) in their lives, there is no shared convention about its “meaning”—namely, what it is used for. They can rely only on its inherent characteristics as clues to deduce its possible uses, without influence from cultural conventions or practice. These characteristics are pure ecological cognitive affordances, the inherent, raw, Gibson-style affordances of an object in its environment.

Their ecological exploration led to interesting and sometimes amusing effects. One guy got his finger stuck in the top, and the children thought *that* was funny. By blowing over the opening, some discovered it could be an entertainment device with which they could make odd sounds and music. After much exploration they saw the smooth, hard, and curved surface was *perfect* for curing snake skins. Perhaps it could even be used as a weapon to attack one another.

Every day they discovered a new use for the thing sent by the gods. One obvious use was to transport water. Its efficacy in mashing soft roots and other vegetables and grinding grain made it a most useful labor-saving device. You can learn quite a bit about affordances by watching Kalahari Bushmen explore a newfound Coke bottle.

That these affordances became so apparent to the Bushmen, but might not be obvious to most people from an industrialized part of the world, indicates the impact of social experience and cultural conventions on influencing, and even prejudicing, one’s perception of an object’s affordances.

Gibson’s ecological view of affordances

Perspective on affordances that says knowledge inherent in objects and devices can give us clues as to their operation. Gibson (1977) studied the relationship between a living being (e.g., a human) and its environment, in particular what the environment offers or affords being—for example, how a horizontal rigid surface affords support for a person for standing (Section 30.2.1.3).

33.5 HOW FUNCTIONAL AFFORDANCES FIT IN WITH GIBSON’S ECOLOGICAL VIEW

We bring Gibson’s ecological view into UX design by describing a purpose for each instance of a user interaction, namely, the associated functional affordance. Putting the user’s purpose into the picture harmonizes nicely with our interaction- and user-oriented view in which *an affordance helps or aids the user in doing something*.

McGrenere and Ho (2000) also refer to the concept of application usefulness, something they call “affordances in software,” which are at the root of supporting a connection between the dual concepts of usability and

usefulness (Landauer, 1995). In an external view, it is easy to see a system function as an affordance because it helps the user do something in the work domain.

This again demonstrates the need for a richer vocabulary, and conceptual framework, to take the discussion of affordances beyond user interfaces to the larger context of overall system design. We use the term *functional affordance* to denote this kind of higher level user enablement in the work domain.

Usefulness

A component of user experience based on utility, system functionality that gives users the ability to accomplish the goals of work (or play) through using the system or product (Section 1.4.3).

33.6 WHERE DID UX DESIGN GUIDELINES COME FROM?

We cannot talk about UX design guidelines without giving a profound acknowledgement to what is perhaps the mother (and father) of all guidelines publications, the book of 944 design guidelines for text-based user interfaces of bygone days that Smith and Mosier of Mitre Corporation developed for the U.S. Air Force (Mosier & Smith, 1986; Smith & Mosier, 1986).

We were already working in human-computer interaction (HCI) and read it with great interest when it came out. Almost a decade later, an electronic version became available (Iannella, 1995). Other early guidelines collections include Engel and Granda (1975), Brown (1988), and Boff and Lincoln (1988).

UX design guidelines appropriate to the technology of the day appeared throughout the history of HCI, including “the design of idiot-proof interactive programs” (Wasserman, 1973); ground rules for a “well-behaved” system (Kennedy, 1974); design guidelines for interactive systems (Pew & Rollins, 1975); usability maxims (Lund, 1997); and eight golden rules of interface design (Shneiderman, 1998). Every author and practitioner had a favorite set of design guidelines or maxims.

Eventually, of course, the attention of design guidelines followed the transition to graphical user interfaces (Nielsen, 1990; Nielsen et al., 1992). As GUIs evolved, many of the guidelines became platform specific, such as style guides for Microsoft Windows and Apple. Each has its own set of detailed requirements for compliance with the respective product lines.

As an example from the 1990s, an interactive product from Apple called Making it Macintosh (Alben, Faris, & Saddler, 1994; Apple Computer Inc, 1993) used computer animations to highlight the Macintosh user interface design principles, primarily to preserve the Macintosh look and feel. Many of the early style guides, such as OSF Motif (Open Software Foundation, 1990) and IBM’s Common User Access (Berry, 1988), came built into software tools for enforcing that particular style.

The principles behind the guidelines came mainly from human psychology. Our friend [Tom Hewett \(1999\)](#) was probably the most steadfast HCI voice for understanding psychology as a foundation for UX design principles and guidelines. These principles first evolved into design guidelines in human factors engineering.

Some UX design guidelines, especially those coming from human factors, are supported with empirical data. Most guidelines, however, have earned their authority from a strong grounding in the practice and shared experience of the UX community—experience in design and evaluation, experience in analyzing and solving UX problems.

Based on the National Cancer Institute’s Research-Based Web Design and Usability Guidelines project begun in March 2000, the U.S. Department of Health and Human Services has published a book containing an extensive set of UX design guidelines and associated reference material ([U.S. Department of Health and Human Services, 2006](#)). Each guideline has undergone extensive internal and external review with respect to tracking down its sources, estimating its relative importance in application, and determining the “strength of evidence,” for example, strong research support versus weak research support, supporting the guideline.

As is the case in most domains, design guidelines finally opened the way for standards ([Abernethy, 1993](#); [Billingsley, 1993](#); [Brown, 1993](#); [Quesenbery, 2005](#); [Strijland, 1993](#)).